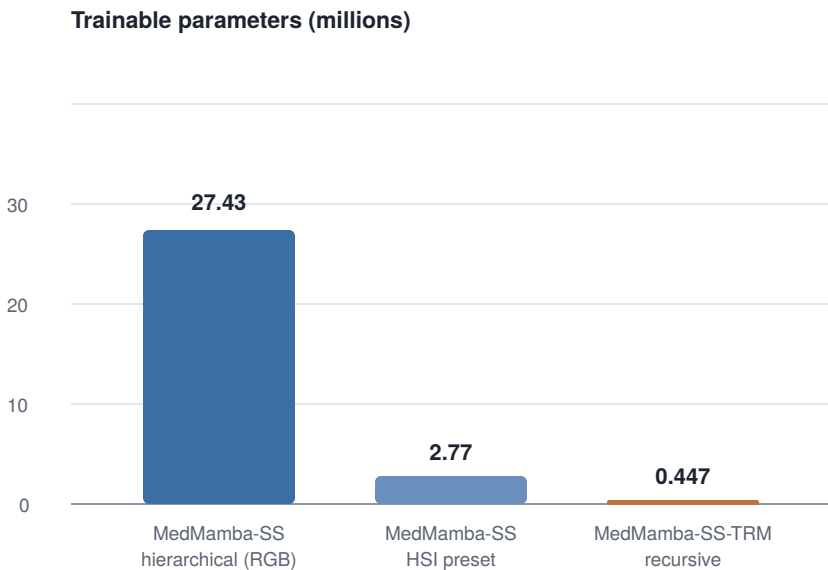


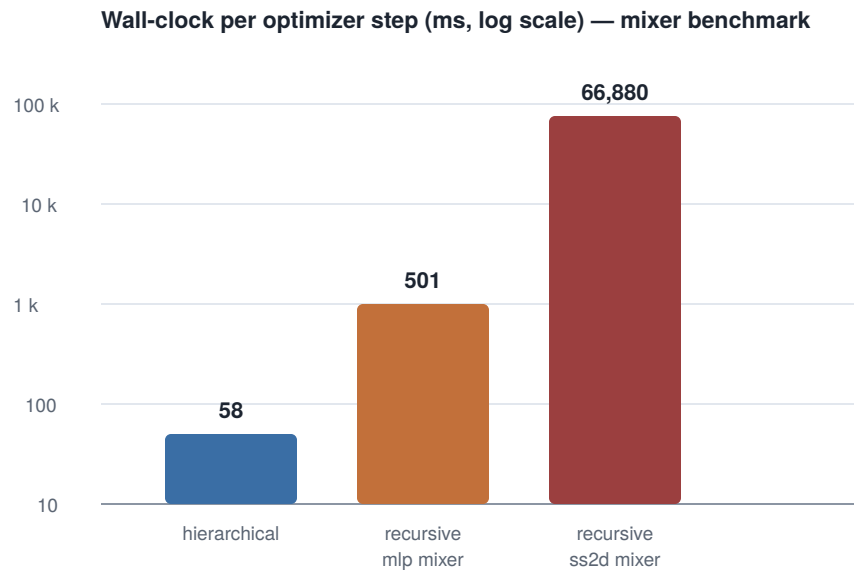
**Fig. 4 — The trade-off measured: parameters fall 6.2× on HSI and 61.4× on PAD, while arithmetic rises 22×**



**Same task, measured: on 32-band 11×11 HSI the hierarchical model is 2.77 M parameters at 0.278 GFLOPs, the recursive one 0.446 M at 6.203 GFLOPs — 6.2× fewer parameters for 22.3× the arithmetic.**

The 27.43 M bar is the PAD/RGB preset (6 classes, 224<sup>2</sup>, stride 8), where the ratio is 61.4× and the FLOP gap 16.4×.

*Without checkpointing the 0.44 M-parameter model exhausts 16 GB of VRAM — the clearest single statement of what recursion costs.*



Bars 1–2 are like-for-like at the epoch level: ≈5 min vs ≈24 min per epoch over the same 740,800 samples.

Bar 3 is the recursive model with the mixer flag changed and no other fix applied (fp32, reconstruction of a pure-Python selective scan called 168× per step, turning a 25-minute epoch into a measured 92-hour epoch).

*From a controlled single-run sweep; training wall-clock is not quoted, as those runs shared the card.*